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SUBJECT:PYTHON
BATCH:03

Inventory Management System in Python

Using File Integration

Aim

To create a simple inventory management system that can **view, add, and update product quantities** using CSV files for persistent storage.

Procedure

1. Import csv, os, and tempfile modules.
2. Define a CSV file path in a safe temporary directory.
3. **Load inventory** (load_inventory) – read existing products and quantities from the CSV file.
4. **Save inventory** (save_inventory) – write products and quantities to the CSV file.
5. **View inventory** (view_inventory) – display all products and their quantities.
6. **Run predefined commands** (run_commands) – simulate user actions:
    - view → display inventory
    - add → add a new product with quantity
    - update → update quantity of an existing product
    - exit → terminate the program
7. Execute the program with predefined commands to demonstrate functionality.

Code:

```
import csv

import os

import tempfile

# File in a safe temp directory

FILE_NAME = os.path.join(tempfile.gettempdir(), "inventory.csv")
```

```
def load_inventory():
    inventory = {}
    try:
        with open(FILE_NAME, "r", newline="") as file:
            reader = csv.reader(file)
            for row in reader:
                product, qty = row
                inventory[product] = int(qty)
    except FileNotFoundError:
        pass
    return inventory
def save_inventory(inventory):
    with open(FILE_NAME, "w", newline="") as file:
        writer = csv.writer(file)
        for product, qty in inventory.items():
            writer.writerow([product, qty])
def view_inventory(inventory):
    if not inventory:
        print("Inventory is empty!")
    for product, qty in inventory.items():
        print(f"{product}: {qty}")
def run_commands(commands):
    """Run commands from a predefined list instead of input()"""
    inventory = load_inventory()
    for command in commands:
```

```
        action = command[0]
        if action == "view":
            view_inventory(inventory)
        elif action == "add":
            product, qty = command[1], int(command[2])
            inventory[product] = qty
            save_inventory(inventory)
            print(f"Added: {product} ({qty})")
        elif action == "update":
            product, qty = command[1], int(command[2])
            if product in inventory:
                inventory[product] = qty
                save_inventory(inventory)
                print(f"Updated: {product} ({qty})")
            else:
                print(f"{product} not found")
        elif action == "exit":
            print("Exiting...")
            break
if __name__ == "__main__":
    # Predefined commands simulating user input
    commands = [
        ("view",),
        ("add", "Laptop", 5),
        ("add", "Mouse", 20),
```

```
        ("view",),
        ("update", "Laptop", 8),
        ("view",),
        ("exit",)
    ]
    print(f"Inventory file path: {FILE_NAME}\n")
    run_commands(commands)
```

Output:

Output

```
Inventory is empty!
Added: Laptop (5)
Added: Mouse (20)
Laptop: 5
Mouse: 20
Updated: Laptop (8)
Laptop: 8
Mouse: 20
Exiting...
```

Result

The program added "Laptop" (5) and "Mouse" (20), displayed the inventory, updated "Laptop" to 8, and saved all changes persistently in the CSV file.

Conclusion

The inventory management system successfully handles viewing, adding, and updating products using CSV file storage, ensuring data persistence and easy management.